

WEEK 1 ASSIGNMENTS

1) File: Design Patterns and Principles

Exercise: Exercise 1: Implementing the Singleton Pattern

Code for the exercise:

```
class logger{
    private static logger instance;
    private logger(){
        System.out.println("Logger instance created");
    }
    public static logger
        getInstance(){ if(instance==null){
            instance= new logger();
        }
        return instance;
    }
    public void log(String m){
        System.out.println("Log: "+m);
    }
}

public class SingletonPatternExample
{ public static void main(String[] args) {
    logger logger1= logger.getInstance();
    logger1.log("Application Created");

    logger logger2= logger.getInstance();
    logger2.log("User Logged in");

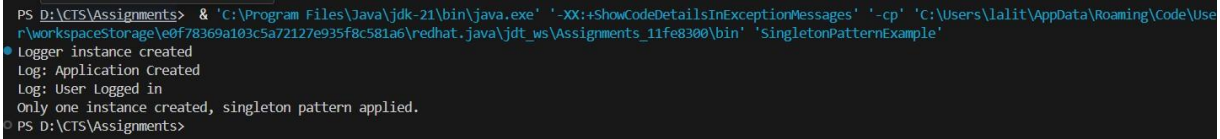
    if(logger1==logger2){
        System.out.println("Only one instance created, singleton pattern applied.");
    }
}
```

```

    }
    else{
        System.out.println("More than one instance found, singleton pattern not applied");
    }
}
}
}

```

Output screenshot:



```

PS D:\CTS\Assignments> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\lalit\AppData\Roaming\Code\Use\r\workspaceStorage\0f78369a103c5a72127e935f8c581a6\redhat.java\jdt_ws\Assignments_11fe8300\bin' 'SingletonPatternExample'
● Logger instance created
Log: Application Created
Log: User Logged in
Only one instance created, singleton pattern applied.
PS D:\CTS\Assignments>

```

2) File: Design Patterns and Principles

Exercise: **Exercise 2: Implementing the Factory Method Pattern**

Code for the exercise:

➤ Document.java

```

package FactoryMethodPatternExample;

public interface Document{

    void open();

}

```

➤ DocumentFactory.java

```

package FactoryMethodPatternExample;

public abstract class DocumentFactory {

    public abstract Document createDocument();

}

```

➤ ExcelDocument.java

```

package FactoryMethodPatternExample;

public class ExcelDocument implements Document

{ @Override
    public void open() {
        System.out.println("Opening Excel document...");
    }

}

```

➤ ExcelFactory.java

```

package FactoryMethodPatternExample;

```

```
public class ExcelFactory extends DocumentFactory
{ @Override
  public Document createDocument()
  { return new ExcelDocument();
  }
}
```

➤ PdfDocument.java

```
package FactoryMethodPatternExample;
public class PdfDocument implements Document
{ @Override
  public void open() {
    System.out.println("Opening PDF document...");
  }
}
```

➤ PdfFactory.java

```
package FactoryMethodPatternExample;

public class PdfFactory extends DocumentFactory
{ @Override
  public Document createDocument()
  { return new PdfDocument();
  }
}
```

➤ WordDocument.java

```
package FactoryMethodPatternExample;

public class WordDocument implements Document
{ @Override
  public void open() {
    System.out.println("Opening Word document...");
  }
}
```

➤ WordFactory.java

```
package FactoryMethodPatternExample;
public class WordFactory extends DocumentFactory
{ @Override
  public Document createDocument()
  { return new WordDocument();
  }
}
```

➤ Main.java

```
package FactoryMethodPatternExample;
```

```

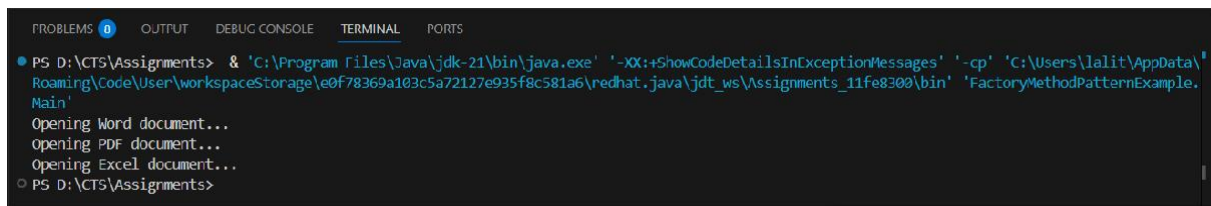
public class Main {
    public static void main(String[] args) {
        // Word
        DocumentFactory wordFactory = new WordFactory();
        Document wordDoc = wordFactory.createDocument();
        wordDoc.open();

        // PDF
        DocumentFactory pdfFactory = new PdfFactory();
        Document pdfDoc = pdfFactory.createDocument();
        pdfDoc.open();

        // Excel
        DocumentFactory excelFactory = new ExcelFactory();
        Document excelDoc = excelFactory.createDocument();
        excelDoc.open();
    }
}

```

Output Screenshots:



```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\CTS\Assignments> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\lalit\AppData\Roaming\Code\User\workspaceStorage\e0f78369a103c5a72127e935f8c581a6\redhat.java\jdt_ws\Assignments_11fe8300\bin' 'FactoryMethodPatternExample.Main'
Opening Word document...
Opening PDF document...
Opening Excel document...
PS D:\CTS\Assignments>

```

3) File: Algorithms Data Structures

Exercise: Exercise 2: E-commerce Platform Search Function

Code for the exercise:

```

import java.util.*;

class product{
    int productId;
    String name;
    String cat;
    public product(int p, String n, String
        c){ this.productId=p;
        this.name=n;
        this.cat=c;
    }
}

public class BinarySearch{
    public static product binarySearch(product[] pro, String ta){
        Arrays.sort(pro, Comparator.comparing(p -> p.name.toLowerCase()));
    }
}

```

```

int left=0;
int right= pro.length-1;
while(left<=right){
    int mid= (left+right)/2;
    int com= pro[mid].name.compareToIgnoreCase(ta);

    if(com==0){
        return pro[mid];
    }
    else
        if(com<0){ le
            ft= mid+1;
        }
        else{
            right=mid-1;
        }
    }
return null;
}

public static void main(String[] args)
{ product[] catalog = {
    new product(1, "Laptop", "Electronics"),
    new product(2, "Shirt", "Clothing"),
    new product(3, "Phone", "Electronics"),
    new product(4, "Book", "Education"),
    new product(5, "Watch", "Accessories")
};
product cat= binarySearch(catalog, "phone");
if(cat!=null){
    System.out.println("Binary Search: "+"Product Id: "+cat.productId+", Product name:
        "+cat.name+", Category: "+cat.cat);
}
else{
    System.out.println("Binary Search: Not found");
}
}
}

```

Output Screenshots:



```

PROBLEMS 8 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\CTS\Assignments> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\lalit\AppData\Roaming\Code\User\workspaceStorage\e0f78369a103c5a72127e935f8c581a6\redhat.java\jdt_ws\Assignments_11fe8300\bin' 'BinarySearch'
Binary Search: Product Id: 3, Product name: Phone, Category: Electronics
PS D:\CTS\Assignments>

```

4) File: Algorithms Data Structures

Exercise: Exercise 7: Financial Forecasting

Code for the exercise:

```
public class FinancialForecast {  
    static double futVal(double CV, double GR, double  
        y){ if(y==0){  
            return CV;  
        }  
        return futVal(CV, GR, y-1)*(1+GR);  
    }  
    public static void main(String[] args)  
    { double c= 10000;  
      double g=0.08;  
      double y=5;  
      double F= futVal(c, g, y);  
      System.out.printf("Future Value: %.2f", F);  
    }  
}
```

Output Screenshots:



```
PROBLEMS 8 OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS D:\CTS\Assignments> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\lalit\AppData\Roaming\Code\User\workspaceStorage\e0f78369a103c5a72127e935f8c581a6\redhat.java\jdt_ws\Assignments_11fe8300\bin' 'FinancialForecast'  
Future Value: 14693.28  
PS D:\CTS\Assignments>
```